

College of Computer and Information Sciences
Department of Information Systems

modern application development IS324

KSUPay

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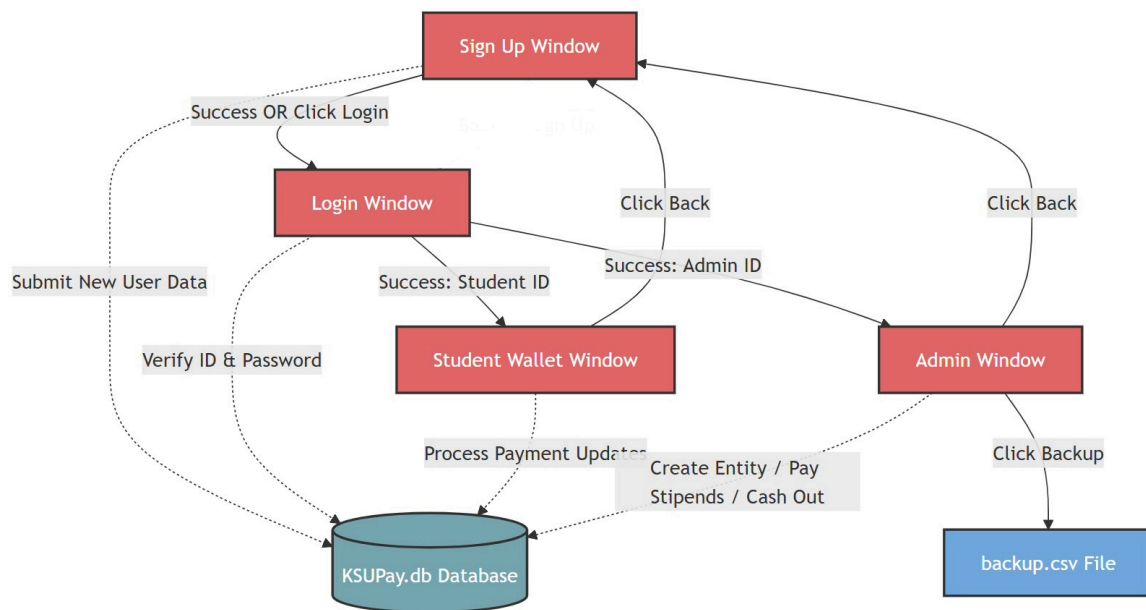
introduction

KSUPay is a newly developed payment system designed for King Saud University administration to manage and streamline financial transactions within the campus. The system facilitates various payments, such as purchasing from the bookstore, paying for accommodation, and depositing monthly student stipends. Developed using Python (Tkinter and SQLite), the application features a user-friendly graphical interface with secure login mechanisms and multiple windows tailored for both students and system administrators, ensuring efficient and accurate record-keeping of all campus transactions.

Challenges

- **Challenge:** manage the admin features and make sure everything worked correctly with the database.
- **Solution:** separating each task into its own function and testing them one by one
- **Challenge:** old windows were staying open when moving between screens (Sign-up, Login, and Wallet), making the program messy
- **Solution:** To maintain a clean interface, we used the `destroy()` method to close the current window before opening the next.
- **Challenge:** Preventing money loss if a student accidentally enters a wrong or non-existent target wallet number during a payment
- **Solution:** In the `process_payment` method, we added a validation step (if not `target_data:`) to verify that the target wallet actually exists in the database. If it does not exist, the system safely stops the transaction and alerts the user without deducting any money.
- **Challenge:** Ensuring that the system correctly validates the wallet ID and updates both the student and target balances after the payment.
- **Solution:** I implemented database queries to check if the wallet exists and updated both balances after the transaction, while also handling errors such as invalid wallet ID or insufficient balance.

flow chart



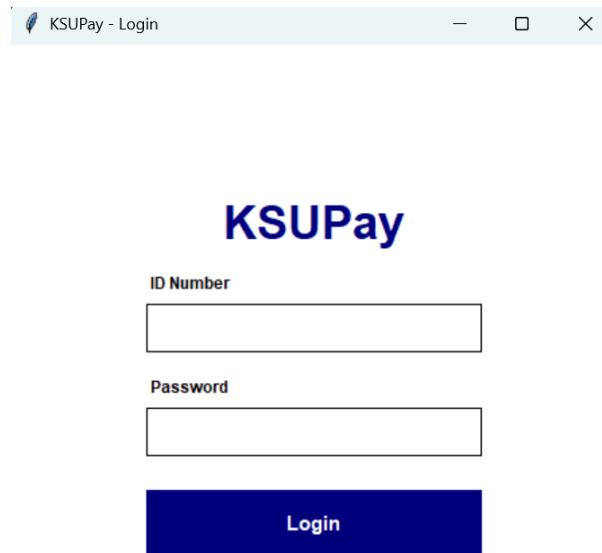
Brief Flowchart Explanation:

This flowchart illustrates the user navigation and data flow of the final KSUPay system:

- **Entry Point (Sign Up):** The application starts here. Users can register new accounts (saving data to the database) or navigate to the Login window.
- **Login & Routing:** Authenticates users via the database and directs them to either the Admin Window (using the specific Admin ID) or the Student Wallet.
- **Student Wallet:** Allows students to process payments, which dynamically updates both sender and receiver balances in the database.
- **Admin Window:** Enables administrators to manage KSU entity wallets, distribute stipends, and export all system records to a backup.csv file.
- **Flow Types:** Solid arrows (-->) represent user navigation between GUI windows, while dotted arrows (-.->) indicate hidden backend operations with the SQLite database.

Screenshots of the GUI

login window



A screenshot of a web browser window titled "KSUPay - Login". The window contains the "KSUPay" logo in blue. Below the logo are two input fields: "ID Number" and "Password". A blue "Login" button is positioned below the password field.

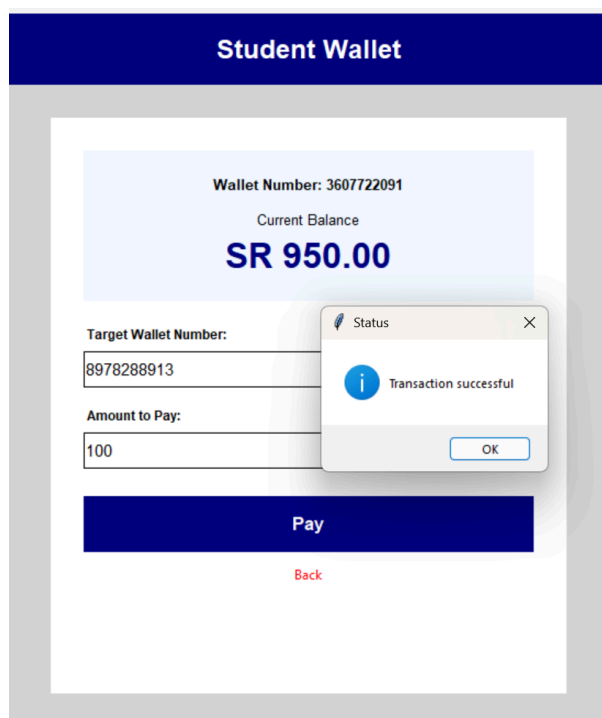
KSUPay

ID Number

Password

Login

student wallet with pay transaction



A screenshot of the "Student Wallet" interface. The header is "Student Wallet". The main content area shows "Wallet Number: 3607722091" and "Current Balance: SR 950.00". Below this are two input fields: "Target Wallet Number:" with the value "8978288913" and "Amount to Pay:" with the value "100". A blue "Pay" button is at the bottom. A modal window titled "Status" is open, displaying a blue information icon and the text "Transaction successful" with an "OK" button. A red "Back" link is visible at the bottom of the main content area.

Student Wallet

Wallet Number: 3607722091

Current Balance

SR 950.00

Target Wallet Number:

8978288913

Amount to Pay:

100

Pay

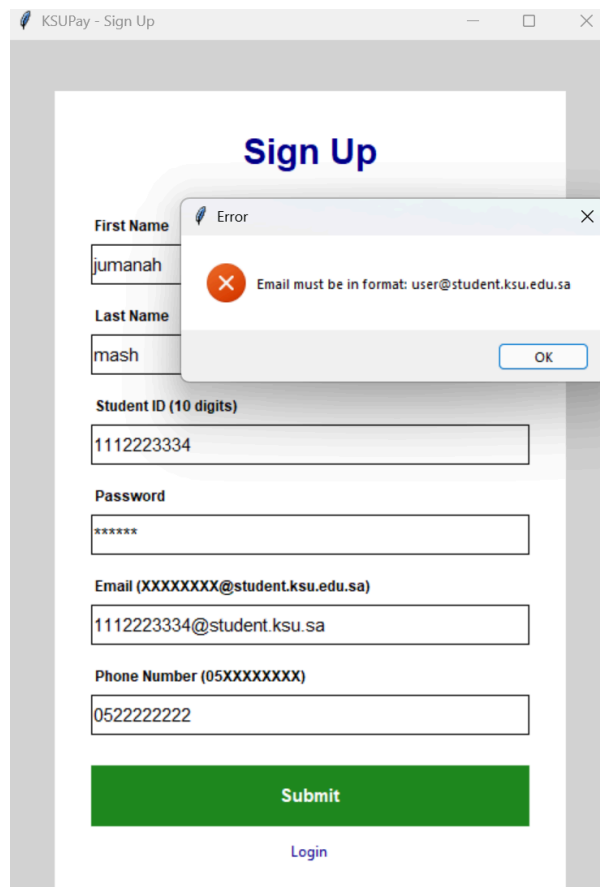
Back

Status

Transaction successful

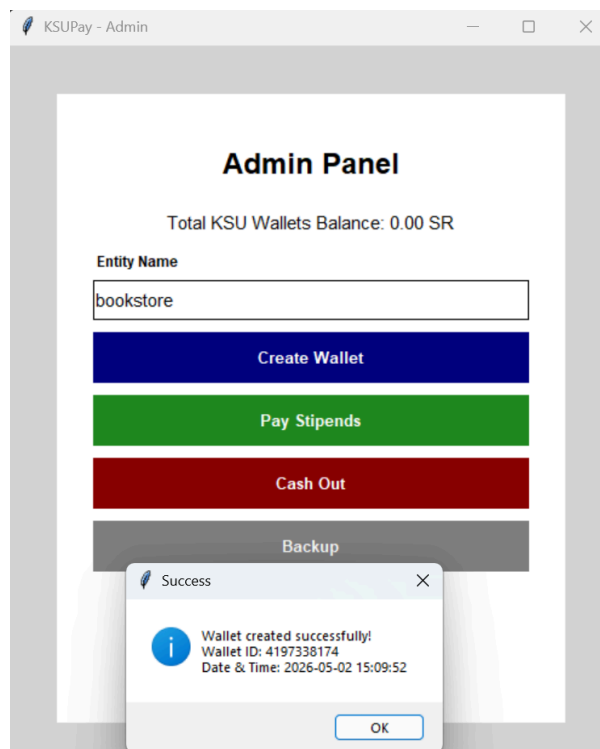
OK

sign up window with error message



The image shows a web browser window titled "KSUPay - Sign Up". The main content area has a "Sign Up" heading. Below it are several input fields: "First Name" (filled with "jumanah"), "Last Name" (filled with "mash"), "Student ID (10 digits)" (filled with "1112223334"), "Password" (filled with "*****"), "Email (XXXXXXXX@student.ksu.edu.sa)" (filled with "1112223334@student.ksu.sa"), and "Phone Number (05XXXXXXXX)" (filled with "0522222222"). At the bottom is a green "Submit" button and a "Login" link. An "Error" dialog box is overlaid on top of the form, displaying a red "X" icon and the message "Email must be in format: user@student.ksu.edu.sa". The dialog box has an "OK" button.

admin window with create wallet transaction



The image shows a web browser window titled "KSUPay - Admin". The main content area has an "Admin Panel" heading. Below it, it displays "Total KSU Wallets Balance: 0.00 SR". There are four buttons: "Create Wallet" (blue), "Pay Stipends" (green), "Cash Out" (red), and "Backup" (grey). A "Success" dialog box is overlaid on top of the buttons, displaying a blue "i" icon and the message "Wallet created successfully! Wallet ID: 4197338174 Date & Time: 2026-05-02 15:09:52". The dialog box has an "OK" button.

contribution of each student

student	contribution
jumanah	database-payment process
Arwa	Login system
Reema	Sign up
Dhai	Admin
Dana	Student Wallet